

Ex No: 1:**Table Creation and Data Insertion:**

SQL

Create table Student(StudentID int PRIMARY KEY, Name varchar(30), Age int);

Create table Courses (CourseID int PRIMARY KEY, CourseName varchar(20));

Create table Enrollments (StudentID int REFERENCES students(student_id), CourseID int REFERENCES courses (course_id));

Insert into Student values(1,'Alice', 20);

Insert into Courses values(101, 'Database Management');

Insert into Enrollments values(1,101);

Simple Queries:

SELECT* FROMStudents;

Output:

StudentID	Name	Age
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1	Alice	20
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2	Bob	22
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3	Charlie	21
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4	David	19
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SELECTName,Age FROM Students WHERE Age>20;

Output:

StudentID	Name	Age
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2	Bob	22
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3	Charlie	21
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Nested Queries:

SELECTNameFROM Students WHERE StudentID IN (SELECT StudentID FROM Enrollments WHERE CourseID = (SELECT CourseID FROM Courses WHERE CourseName = 'Database Management'));

Output:

Name	StudentID
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SELECT CourseID, CourseName FROM Courses WHERE CourseID IN (SELECT CourseID FROM Enrollments GROUP BY CourseID HAVING COUNT(*)>1);

Output: | CourseID | CourseName |

101	Database Management
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103	Web Development
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Subqueries:

SELECTAVG(Age) AS AverageAge FROM Students;

Output:

AverageAge
20.5

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SELECT Name, Age FROM Students WHERE Age > (SELECT AVG(Age) FROM  
Students);
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Output:

Name	Age
Bob	22
Charlie	21